Vacation-rental marketplace — production architecture

Read paths (search, listing) are cached and horizontally scaled; the booking write path is strongly consistent; everything else reacts to events.

Clients

Web app
Next.js · SSR/ISR
listing pages cached at edge

iOS / Android
native apps, same APIs

Host dashboard
listings, calendar, payouts

Edge

CDN + WAF
static assets, HTML cache
bot + DDoS protection

API gateway / BFF
auth (JWT/OIDC), rate limits
GraphQL/REST fan-out
per-screen payloads

Image CDN
resizes photos from object storage, serves AVIF/WebP

Services (Kubernetes, per region)

Search
geo + filters + ranking

Listing
details, photos, amenities

Availability
calendar, holds

Pricing
nightly rates, fees, dynamic pricing

Booking
reservation saga, idempotent requests

Payments
PSP adapters, ledger, payouts

Reviews
ratings, moderation

Messaging
host ↔ guest, websockets

Identity & users
accounts, KYC, wishlists

Media
uploads, variants

Notifications
email · SMS · push

Trust & safety
fraud scoring, holds

Booking write path
hold dates in Redis (10 min TTL) → quote from Pricing → PSP authorise →
insert booking (Postgres exclusion constraint) → capture → BookingConfirmed event

Data & search

PostgreSQL (per service)
bookings: exclusion constraint on date ranges stops double-booking; read replicas

Redis
cache, sessions, availability holds

OpenSearch
geo index, availability bitmaps

Object storage
photos, documents

Payments ledger
append-only, audited

External: PSPs, maps, email/SMS providers
called through adapters with retries + circuit breakers

each service owns its own store; no shared tables

outbox → Kafka

indexes, features, ranking

Events & analytics

Kafka event bus
BookingConfirmed, PriceChanged, ListingUpdated…
outbox pattern from each service

Search indexer (CDC consumers)
keeps OpenSearch in sync within seconds

Data lake + warehouse
S3 + Iceberg → Snowflake/BigQuery

ML platform
search ranking, pricing, fraud models

Deployment, scaling & operations

Multi-region Kubernetes
HPA on CPU/RPS, 3 AZs per region, active-active reads, pinned writes

CI/CD
GitHub Actions → Argo CD, canary + feature flags, IaC (Terraform)

Observability
OpenTelemetry traces, Prometheus, SLOs on search p95 and booking success

Security
secrets in KMS/Vault, PCI scope isolated to Payments, least-privilege IAM